

Comprehensive Deep Research into the "SEF-GRAM" Architecture: Variational World Models with Exact Continuous-Time Latent Dynamics and Critic-Free Multi-Objective RL

The design of resource-efficient planning agents capable of operating within complex, structured environments represents a key challenge in machine learning. Classical model-based reinforcement learning systems rely heavily on reconstructing high-dimensional observation spaces to stabilize their latent transitions, creating a massive computational and memory bottleneck. Autoregressive sequence models attempt to solve planning tasks via sequential, token-level token generation, but they are bounded by quadratic complexity over long contexts and lack an explicit model of spatial or logical world dynamics.

The Stochastic Error-Free Generative Recursive Reasoning Model (SEF-GRAM) represents a novel paradigm that performs complex structured planning entirely within a compact, variational latent space without high-dimensional observation reconstruction. This architecture integrates a Variational Joint Embedding Predictive Architecture (VJEPA) with Error-Free Linear Attention (EFLA) dynamics, a Bayesian Joint Embedding Predictive Architecture (BJEPA) planning framework, and a highly efficient optimization and verification stack. This report provides a state-of-the-art literature review, mathematical validation, and scalability analysis of these integrated components.

Domain 1: Exact Continuous-Time Latent Propagation (EFLA + VJEPA)

Mathematical Formulation of Continuous-Time Latent Dynamics

Traditional recurrent linear attention mechanisms and state-space models operate as discrete-time approximations of an underlying continuous-time dynamical system.¹ This approximation introduces local truncation errors that compound over extended sequence lengths, resulting in representation drift, numerical instability, and a loss of memory precision.¹ Error-Free Linear Attention (EFLA) resolves these limitations by reinterpreting the fast-weight memory update as a continuous-time ordinary differential equation (ODE) and solving it exactly

in closed form, which corresponds to the infinite-order Runge-Kutta ($RK - \infty$) limit.¹

Let $S(t) \in \mathbb{R}^{d \times d}$ be the associative fast-weight state matrix at continuous time t .³ The online memory update, modeled as a continuous learning process with a data-dependent

forgetting mechanism, is governed by the linear matrix ODE ¹:

$$\frac{dS(t)}{dt} = -A(t)S(t) + B(t)$$

where the decay operator $A(t) \in \mathbb{R}^{d \times d}$ and the input injection operator $B(t) \in \mathbb{R}^{d \times d}$ are parameterized by the continuous-time key vector $k_t \in \mathbb{R}^d$, value vector $v_t \in \mathbb{R}^d$, and step size $\beta_t \in \mathbb{R}^+$ ¹:

$$A(t) = \beta_t k_t k_t^\top, \quad B(t) = \beta_t k_t v_t^\top$$

Under a zero-order hold (ZOH) assumption across a discrete time step interval of duration β_t , the exact analytical solution to this system is formulated as ²:

$$S_t = e^{-\beta_t A_t} S_{t-1} + \int_0^{\beta_t} e^{-(\beta_t - \tau) A_t} B_t d\tau$$

Because the transition matrix $A_t = k_t k_t^\top$ is a rank-1 outer product, its spectrum is defined by a single non-zero eigenvalue $\lambda_t = k_t^\top k_t = \|k_t\|_2^2$ associated with the eigenvector k_t . ¹ Exploiting this rank-1 structure, the matrix exponential can be simplified exactly using the projector expansion:

$$e^{-\beta_t A_t} = e^{-\beta_t k_t k_t^\top} = I - \left(\frac{1 - e^{-\beta_t \lambda_t}}{\lambda_t} \right) k_t k_t^\top$$

Letting $\alpha_t = \frac{1 - e^{-\beta_t \lambda_t}}{\lambda_t}$ define the continuous-time decay coefficient ¹, the convolution integral term likewise simplifies to:

$$\int_0^{\beta_t} e^{-(\beta_t - \tau) A_t} B_t d\tau = \int_0^{\beta_t} \left(I - \frac{1 - e^{-(\beta_t - \tau) \lambda_t}}{\lambda_t} k_t k_t^\top \right) \beta_t k_t v_t^\top d\tau = \alpha_t k_t v_t^\top$$

Thus, the exact, error-free update equation for the EFLA state matrix at time t is formulated as ¹:

$$S_t = (I - \alpha_t k_t k_t^\top) S_{t-1} + \alpha_t k_t v_t^\top$$

Within the Variational Joint Embedding Predictive Architecture (VJEPA), the latent state

$z_t \in \mathbb{R}^d$ represents a probabilistic belief state parameterized as a Gaussian distribution $z_t \sim \mathcal{N}(\mu_t, \Sigma_t)$.⁵ The transition dynamics are mapped through the linear fast-weight transition matrix S_t with heteroscedastic, state-dependent process noise covariance $Q_t \in \mathbb{R}^{d \times d}$.⁶

$$z_t = S_t^\top z_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q_t)$$

Because S_t is computed analytically without discretization error, propagating the Gaussian mean μ_{t-1} and covariance Σ_{t-1} of the latent state to the next planning step is modeled as:

$$\mu_t = \mathbb{E} = S_t^\top \mu_{t-1}$$

$$\Sigma_t = \text{Cov}(S_t^\top z_{t-1} + \epsilon_t) = S_t^\top \Sigma_{t-1} S_t + Q_t$$

This formulation guarantees that the latent belief state is propagated along a continuous-time manifold without accumulating the local truncation errors of $O(\beta_t^2)$ that typically affect discrete Euler-based recurrent updates.¹

Architectural Paradigm Comparison

To understand the positioning of EFLA within sequence modeling, it is necessary to compare its dynamics with traditional state-estimation and modern state-space architectures.

- **Kalman Filtering:** While Kalman filters derive optimal linear state updates, they rely on pre-specified, static transition matrices. EFLA dynamically programs its transition operator S_t at every step as a function of incoming keys and values, merging the mathematical precision of linear systems with the representation capacity of data-dependent fast-weight programmers.¹
- **Predictive State Representations (PSRs):** PSRs represent dynamical systems through predictions of future observables. VJEPA unifies representation learning with PSRs by demonstrating that the learned latent state serves as a sufficient information state for optimal control without observation reconstruction, while bypassing the complex spectral learning algorithms typically required by classic PSRs.⁶
- **Modern State Space Models (Mamba, Mamba-2, S4):** Modern SSMs restrict the transition matrix to a diagonal or low-rank structure to maintain linear computational complexity $O(Ld)$.⁹ For example, Mamba-2 utilizes structured state-space operators where the decay is strictly scalar or channel-wise diagonal.¹⁰ EFLA, conversely, supports

a full matrix-valued state $S_t \in \mathbb{R}^{d \times d}$ ³, which allows for cross-channel interactions and complex associative mapping. Despite this increased expressivity, EFLA maintains a linear-time complexity of $O(Ld^2)$ by restricting the dynamic update step to a rank-1 outer product, ensuring high-fidelity continuous-time tracking without the quadratic bottleneck of softmax attention.¹

- **Linear Recurrence Evolutions:** The development of linear recurrences has progressed from early linear attention to the gated delta rule of Gated DeltaNet and Kimi Delta Attention (KDA).¹⁰ Gated DeltaNet-2 decouples the erase and write gates on the key and value axes to prevent memory scrambling.¹⁰ While these models rely on empirical gating mechanisms, EFLA derives its gating structures directly from the continuous-time physical transition of the system, establishing a theoretical bridge between associative memories and continuous-time dynamical systems.¹

Numerical Instabilities and Stabilization Pipeline

When evaluating the continuous decay term $\alpha_t = \frac{1 - e^{-\beta_t \lambda_t}}{\lambda_t}$ in low-precision floating-point formats (such as IEEE fp16 or bf16), the system is highly susceptible to two numerical failure modes.³

Spectral Radius Divergence

The transition operator is governed by the matrix $M_t = I - \alpha_t k_t k_t^\top$.² The eigenvalue of M_t in the direction of k_t is $1 - \alpha_t \lambda_t = e^{-\beta_t \lambda_t}$.² For all vectors orthogonal to k_t , the eigenvalue is exactly 1.¹³ While the continuous-time formulation guarantees that $e^{-\beta_t \lambda_t} \in (0, 1]$ (marginal stability)², a negative step size ($\beta_t < 0$) or severe weight drift can push the spectral radius $\rho(M_t) > 1$, causing exponential divergence of the latent state. $\lambda_t \rightarrow 0$

Numerical Underflow and Singularity at

When the key norm is small (e.g., during network initialization, padding tokens, or dropout), the eigenvalue $\lambda_t = \|k_t\|_2^2$ approaches zero.² Computing α_t directly leads to a $0/0$ indeterminate form, triggering severe floating-point division-by-zero errors or catastrophic loss of precision (subnormal numbers).

To guarantee numerical stabilization across long contexts, the transition operator is wrapped in a robust stabilization pipeline:

```

Python
import torch

def stable_efla_alpha(beta: torch.Tensor, key: torch.Tensor, eps: float = 1e-5) -> torch.Tensor:
    # Compute the eigenvalue lambda (norm squared of key)
    lambda_1 = torch.sum(key * key, dim=-1, keepdim=True) # Shape: (B, L, 1)
    beta_lambda = beta * lambda_1

    # 1. expm1 numerical stabilization for small or near-zero eigenvalues
    # Underflow/Singularity avoidance via Taylor expansion when beta * lambda is small
    taylor_mask = (beta_lambda < eps)

    # Taylor series approximation of (1 - exp(-x)) / x = 1 - x/2 + x^2/6
    alpha_taylor = beta * 0.5 * beta_lambda + 1.0 / 6.0 * beta * beta_lambda * beta_lambda

    # Direct computation using expm1 for numerically stable regimes
    alpha_direct = -torch.expm1(-beta_lambda) / torch.clamp(lambda_1, min=1e-8)

    alpha = torch.where(taylor_mask, alpha_taylor, alpha_direct)
    return alpha

```

Furthermore, key vectors are strictly regularized using sequence-max normalization to prevent explosive magnitude scaling ¹⁴:

$$k_t \leftarrow \frac{k_t}{\max_{s \leq T} \|k_s\|_2 + \epsilon}$$

Domain 2: Manifold Alignment and Exposure Bias in Latent World Models

Exposure Bias and Representation Drift in Latent-Space Rollouts

Exposure bias arises from the fundamental mismatch between the training and inference regimes of autoregressive or recurrent world models.¹⁵ During self-supervised training, the transition model is optimized under a "teacher-forcing" regime where the input context at step t

is the encoded representation of the ground-truth observation $z_{t-1} = f_{\theta}(x_{t-1})$.¹⁷ During planning or closed-loop evaluation, however, the model must condition its predictions on its own previously generated (and therefore noisy) latent states:

$$\hat{z}_t \sim p_{\phi}(Z_t | \hat{z}_{t-1}, u_{t-1})$$

Because the model is never exposed to its own predictive distribution during standard training, minor single-step prediction errors $e_t = \hat{z}_t - z_t$ accumulate.¹⁷ Over a planning horizon H , these errors compound, dragging the predicted trajectory off the valid latent manifold mapped by the encoder.¹⁶ This shift, known as *representation drift*, manifests as a loss of metric properties, dimensional collapse, and eventual disintegration of the topological structure of the latent space, rendering long-horizon planning impossible.¹⁶

Evaluation of Alignment Mechanisms

To prevent representation drift and preserve the structural integrity of the latent space during closed-loop planning, several self-supervised alignment paradigms are evaluated in the table below:

Alignment Paradigm	Mathematical Objective	Computational Overhead	Topological & Metric Preservation	Stability Against Collapse
Contrastive Predictive Coding (CPC)	InfoNCE loss maximizing mutual information via negative sampling ²⁰	High; requires a large memory buffer for negative samples ²⁰	Poor; forces representations to a hypersphere, discarding metric distances	Moderate; vulnerable to sampling bias and representation collapse
VICReg (Variance-Covariance Regularization)	Hinge loss on variance and covariance constraints to maintain dimension variance ²¹	Low; $O(d^2)$ complexity on batch covariance ²¹	Moderate; decorrelates latent dimensions but lacks explicit spatial geometry mapping	High; mathematically guarantees non-collapse without negative sampling

Latent Adversarial Training	Minimax objective using a discriminator to align prediction and target densities	High; requires training an auxiliary discriminator network	Poor; aligns probability distributions but warps local metric topology	Low; highly susceptible to mode collapse and training oscillations
Self Forcing (with KV-Caching)	Video-level distribution matching loss applied over full autoregressive self-rollouts ¹⁸	High; requires gradient backpropagation through time over rollouts ¹⁸	Moderate; enforces temporal consistency but lacks explicit metric alignment	High; directly aligns training with the inference rollout distribution ²²
Backwards Aggregation (BAGger)	DAGger-style aggregation that reverses self-rollouts to learn corrective trajectories ¹⁶	Moderate; requires running a secondary corrective optimization loop ¹⁶	Moderate; explicitly corrects drift vectors back to the true manifold ¹⁶	High; robustly handles out-of-distribution transitions by training on errors
Theoretical-Latent Hypothesis Encoding (TLHE)	Pairwise Metric Latent Loss comparing EFLA predictions with target-encoder projections ⁵	Low; direct latent projection without high-dimensional decoding ⁵	Strict; preserves global and local geometry via physical distance mapping ²³	Complete; prevented via EMA Target Encoder and variational ELBO objectives ⁵

Pairwise Distance Optimization and Topological Preservation

To guarantee that the latent manifold constructed by SEF-GRAM is topologically and metrically faithful to the physical environment (e.g., preserving Euclidean distances in a physical maze or state transitions in combinatorial games), the latent space is constrained using a **Metric Latent**

Loss (MLL).²³ Let $X = \{x_1, \dots, x_B\}$ represent a batch of physical states, and let $Z = \{z_1, \dots, z_B\}$ be their corresponding latent encodings generated by the online encoder $z_i = f_\theta(x_i)$.²³ Let $d_X(x_i, x_j)$ define the true geodesic or physical distance

between states, and let $d_Z(z_i, z_j) = \|z_i - z_j\|_2$ define the Euclidean distance in the continuous latent manifold.²³ The MLL is formulated as:

$$\mathcal{R}_{\text{MLL}} = \frac{1}{B^2} \sum_{i=1}^B \sum_{j=1}^B (d_Z(z_i, z_j) - d_X(x_i, x_j))^2$$

This loss is analyzed using the Latent Metric Space (LMS) model and Manifold-Matching Autoencoders (MMAE) to establish rigorous topological guarantees.²³

Theorem: Isometric and Homeomorphic Embedding of the Latent Space

Let $(\mathcal{X}, d_X)$ be a compact physical metric space, and let $f_\theta : \mathcal{X} \rightarrow \mathcal{Z} \subset \mathbb{R}^d$ be a continuous neural network encoder. If the Metric Latent Loss converges to zero ($\mathcal{R}_{\text{MLL}} \rightarrow 0$), then:

1. **Metric Isometry:** The mapping f_θ is a global isometry between the physical space $(\mathcal{X}, d_X)$ and the latent manifold $(\mathcal{Z}, \|\cdot\|_2)$.²⁴ That is:

$$\forall x_i, x_j \in \mathcal{X}, \quad \|f_\theta(x_i) - f_\theta(x_j)\|_2 = d_X(x_i, x_j)$$

2. **Topological Homeomorphism:** Because $(\mathcal{X}, d_X)$ is a compact metric space, any continuous, injective mapping to a metric space is a homeomorphism onto its image.²⁴ Since a global isometry is strictly injective ($f_\theta(x_i) = f_\theta(x_j) \iff \|f_\theta(x_i) - f_\theta(x_j)\|_2 = 0 \iff d_X(x_i, x_j) = 0 \iff x_i = x_j$), the learned latent space $\mathcal{Z}$ is topologically equivalent (homeomorphic) to the physical space $\mathcal{X}$.²⁴

This topological guarantee ensures that continuous paths planned in the latent space $\mathcal{Z}$ correspond to physically realizable, continuous paths in the environment $\mathcal{X}$, preventing the planner from passing through obstacles or violating topological boundaries.²⁷

Domain 3: Zero-Shot Planning via Product of Experts (BJEPA PoE)

Bayesian JEPA Factorization and Product of Experts

The Bayesian Joint Embedding Predictive Architecture (BJEPA) formalizes latent-space planning as probabilistic inference over future latent states.⁶ It decomposes the predictive

distribution of a future latent Z_t into two distinct, modular sources of information: a learned, data-driven *dynamics model* (representing physical transition constraints) and a *prior model* (representing task-specific objectives or safety boundaries) ⁶:

$$p_{\text{predictive}}(Z_t|Z_{t-1}, \eta) \propto p_{\text{like}}(Z_t|Z_{t-1}) \cdot p_{\text{prior}}(Z_t|\eta)$$

where η parameters describe task-specific goals or constraints. ⁶

Assuming both experts are modeled as Gaussian distributions, the dynamics likelihood is

defined by the EFLA transition output $p_{\text{like}}(Z_t|Z_{t-1}) = \mathcal{N}(Z_t; \mu_{\text{dyn}}, \Sigma_{\text{dyn}})$ ⁶, and the task-specific prior is defined by an energy-based target field

$p_{\text{prior}}(Z_t|\eta) = \mathcal{N}(Z_t; \mu_{\text{prior}}, \Sigma_{\text{prior}})$ ⁶. The joint predictive belief is a Product of Experts (PoE), which can be solved analytically via precision-weighted addition ²⁹:

$$\Sigma_{\text{post}} = \left(\Sigma_{\text{dyn}}^{-1} + \Sigma_{\text{prior}}^{-1} \right)^{-1}$$

$$\mu_{\text{post}} = \Sigma_{\text{post}} \left(\Sigma_{\text{dyn}}^{-1} \mu_{\text{dyn}} + \Sigma_{\text{prior}}^{-1} \mu_{\text{prior}} \right)$$

This analytical formulation allows for zero-shot task transfer: the underlying physical dynamics model (p_{like}) remains completely frozen, while the prior expert (p_{prior}) is swapped or modified dynamically at test-time to represent new goals or constraints. ⁶

Failure Modes of Analytical PoE in Non-Gaussian Environments

The analytical precision-weighted formulation relies on the assumption of unimodal Gaussian densities. ⁶ This assumption breaks down in complex physical environments characterized by non-convex, non-Gaussian, or multi-modal constraints:

1. **Obstacle Avoidance Fields:** A prior representing a physical wall or obstacle must place near-zero probability mass on the obstacle's coordinates, creating a multi-modal "ring" or "split" distribution. Multiplying a Gaussian dynamics prior with a multi-modal obstacle prior yields a non-convex posterior that cannot be represented by a single mean μ_{post} .
2. **Logical Constraints:** Disjunctive constraints (e.g., "reach Goal A OR Goal B") introduce discrete multi-modality. Precision-weighted averaging over such constraints collapses the mean to the uninformative midpoint (which may lie directly inside an obstacle), leading to planning failure.

Non-Gaussian Latent Fusion Techniques

To scale SEF-GRAM to non-convex settings, three alternative latent fusion strategies are

evaluated.

Energy-Based Langevin Dynamics (SGLD)

The posterior distribution is parameterized as an energy-based model:

$$p(Z_t|Z_{t-1}, \eta) \propto \exp(-E_{\text{dyn}}(Z_t, Z_{t-1}) - E_{\text{prior}}(Z_t, \eta))$$

where $E_{\text{dyn}}(Z_t, Z_{t-1}) = \frac{1}{2}(Z_t - \mu_{\text{dyn}})^\top \Sigma_{\text{dyn}}^{-1}(Z_t - \mu_{\text{dyn}})$, and $E_{\text{prior}}(Z_t, \eta)$ is an arbitrary non-convex energy function.⁶ Sampling is performed via Stochastic Gradient Langevin Dynamics (SGLD):

$$Z_t^{(k+1)} = Z_t^{(k)} - \frac{\delta}{2} \nabla_Z (E_{\text{dyn}}(Z_t^{(k)}, Z_{t-1}) + E_{\text{prior}}(Z_t^{(k)}, \eta)) + \sqrt{\delta} \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, I)$$

- *Trade-off*: SGLD handles highly non-convex energy landscapes but is computationally slow, requiring iterative gradient steps inside the planning loop.

Latent Particle Filtering

The belief state is represented by a set of P discrete particles $\{Z_t^{(p)}\}_{p=1}^P$.³² Particles are propagated through the continuous-time transition dynamics $Z_t^{(p)} \sim p_{\text{like}}(Z_t|Z_{t-1}^{(p)})$ and then reweighted based on the prior expert evaluate:

$$w_t^{(p)} \propto \exp(-E_{\text{prior}}(Z_t^{(p)}, \eta))$$

- *Trade-off*: This approach preserves multi-modal hypothesis tracking, but is prone to *particle depletion* as the latent dimensionality d increases, requiring $O(e^d)$ particles to cover the state space.

Latent Cross-Entropy Method (CEM)

A population of candidate action trajectories is sampled, rolled out through the latent dynamics, and evaluated against the prior constraints. The mean and variance of the action distribution are iteratively updated using the top K "elite" trajectories.

- *Trade-off*: CEM is highly parallelizable and robust to non-convex boundaries, but lacks a formal probabilistic treatment of uncertainty and can get trapped in local minima in high-dimensional spaces.

Domain 4: High-Efficiency Training Paradigms (GDPO + Muon + DiffusionBlocks)

Optimization Synergy: Muon Orthogonalization and Policy

Regularization

The SEF-GRAM optimization stack employs a hybrid optimizer combining standard SGD or AdamW with **Momentum Orthogonalized by Newton-Schulz (Muon)**.³³ Muon isolates the 2D spatial weight matrices of the hidden layers and applies a spectral post-processing step to replace the gradient update with its nearest orthogonal matrix.³³ This is achieved using an iterative polynomial approximation of the polar decomposition³⁶:

$$\mathbf{X}_{t+1} = a_t \mathbf{X}_t + b_t \mathbf{X}_t \mathbf{X}_t^\top \mathbf{X}_t + c_t (\mathbf{X}_t \mathbf{X}_t^\top)^2 \mathbf{X}_t$$

For a weight update matrix $G \in \mathbb{R}^{n \times m}$, polar decomposition yields $\text{polar}(G) = UV^\top$

where $G = U\Sigma V^\top$ represents the Singular Value Decomposition (SVD).³⁶

This orthogonalization acts as a powerful regularizer during reinforcement learning training, resolving two dominant failure modes:

1. **Prevention of Representation Collapse:** In joint-embedding architectures, the representation space is highly susceptible to collapsing into a low-dimensional manifold where different inputs map to identical representations.²¹ By forcing the update steps of the transition layers to be strictly orthogonal, Muon ensures that the singular values of the weight matrices are uniformly equal to 1 .³⁴ This preserves the volume of the latent space, ensuring that distinct state transitions remain separated.³⁴
2. **Mitigation of Policy Saturation:** In policy optimization, weight parameters often scale rapidly in specific directions, causing the output distribution (softmax probabilities) to saturate near 0 or 1 . This halts exploration. Muon bounds the spectral radius of the parameter updates ($\|W\|_2 \leq 1$), preventing explosive parameter growth, maintaining a smooth optimization landscape, and stabilizing the policy's exploration entropy.³³

To maximize computational throughput on modern hardware, standard Newton-Schulz is

replaced with **Gram Newton-Schulz**, which operates on the smaller $n \times n$ symmetric Gram matrix to reduce rectangular matrix multiplication FLOPs by up to 50%³⁶:

Python

```
def gram_newton_schulz(X: torch.Tensor, iterations: int = 5) -> torch.Tensor:
    # Assumes X shape is (n, m) with n <= m
    # Normalize input to have spectral norm <= 1
```

```

X_norm = X / (torch.linalg.norm(X, ord='fro') + 1e-7)

# Initialize Gram matrix and scaling operator
R = torch.matmul(X_norm, X_norm.t())
Q = torch.eye(X.shape, device=X.device, dtype=X.dtype)

# Polynomial coefficients for approximating the inverse square root
# Coefficients: a=1.875, b=-1.25, c=0.375
for _ in range(iterations):
    A = 1.875 * torch.eye(R.shape, device=R.device, dtype=R.dtype) - 1.25 * R + 0.375
    torch.matmul(R, R)
    Q = torch.matmul(Q, A)
    R = torch.matmul(A, torch.matmul(R, A))

return torch.matmul(Q, X_norm)

```

Group Reward-Decoupled Policy Optimization (GDPO)

Traditional Group Relative Policy Optimization (GRPO) simplifies reinforcement learning by eliminating the critic network, instead computing advantages relative to a sample group of rollouts.³⁷ However, in multi-objective settings, GRPO naively aggregates all reward components prior to normalization³⁸:

$$r_{\text{sum}}^{(i,j)} = r_{\text{correct}}^{(i,j)} + r_{\text{format}}^{(i,j)} + r_{\text{brevity}}^{(i,j)}$$

Applying group normalization to this scalar sum causes different reward combinations to collapse into identical advantage values, severely degrading the precision of the training signal.³⁸ For example, in a two-reward setting, a rollout with scores $(0, 2)$ (satisfying both correctness and format) is mapped to the same relative advantage as $(0, 1)$ when compared to a baseline of zero, failing to reward simultaneous constraint satisfaction.³⁹

Group reward-Decoupled Policy Optimization (GDPO) resolves this by normalizing each reward dimension independently across the group before aggregation, followed by a stabilizing batch-wise normalization.³⁹

Mathematical Advantage Formulation

For a group of G rollouts, let $r_k^{(i,j)}$ represent the score of the j -th response for the k -th reward objective.³⁹

1. **Group-wise Decoupled Normalization:** Compute the group relative advantage for each reward independently³⁹:

$$A_k^{(i,j)} = \frac{r_k^{(i,j)} - \mu_k^{(i)}}{\sigma_k^{(i)} + \epsilon}$$

$$\text{where } \mu_k^{(i)} = \frac{1}{G} \sum_{j=1}^G r_k^{(i,j)} \text{ and } \sigma_k^{(i)} = \sqrt{\frac{1}{G} \sum_{j=1}^G (r_k^{(i,j)} - \mu_k^{(i)})^2} \quad .^{39}$$

2. **Aggregation:** Sum the independent normalized advantages:

$$A_{\text{sum}}^{(i,j)} = \sum_{k=1}^n A_k^{(i,j)}$$

3. **Batch-wise Advantage Normalization:** Normalize the aggregated advantages across the entire training batch B to stabilize gradient magnitude as the number of objectives scales ³⁹:

$$\hat{A}^{(i,j)} = \frac{A_{\text{sum}}^{(i,j)} - \text{mean}_{\text{Batch}}\{A_{\text{sum}}\}}{\text{std}_{\text{Batch}}\{A_{\text{sum}}\} + \epsilon}$$

The table below contrasts GDPO with alternative multi-objective reinforcement learning optimization algorithms:

Optimization Algorithm	Advantage Collapse Prevention	Critic Dependency	Computational / Memory Complexity	Support for Competitiveness / Pareto Trade-offs
PPO + Pareto-active Actor-Critic	High; optimizes along the Pareto frontier.	High; requires n separate critic networks.	High ($O(n \cdot \text{memory overhead})$).	Excellent; explicitly maps the Pareto-optimal trade-offs.
ALARM (Adaptive Learning Rate)	Moderate; scales loss weights dynamically.	High; relies on value-gradient tracking.	Moderate.	Moderate; prone to oscillations in highly non-stationary environments.

Decoupled Length-Efficient RL (DLER)	Poor; relies on heuristic scalar truncation and merging. ³⁷	None.	Low.	Narrow; limited to trading off length and task accuracy. ³⁷
GRPO (Group Relative)	None; aggregates raw rewards prior to normalization, washing out signal resolution. ³⁸	None; completely critic-free. ³⁸	Very Low ($O(1)$ memory overhead).	Poor; easier objectives (format) frequently suppress harder objectives (correctness). ⁴¹
GDPO (Decoupled Normalization)	Complete; decouples group normalization per reward dimension. ³⁹	None; completely critic-free. ³⁹	Low; adds negligible $O(n \cdot$ vector operations. ³⁹	Strong; preserves reward gradients separately, preventing single-objective dominance. ⁴¹

DiffusionBlocks Integration in Reinforcement Learning Training

DiffusionBlocks interprets sequential layer updates in a Transformer as discretized steps of a continuous-time reverse diffusion process⁴³:

$$\frac{dz_\sigma}{d\sigma} = -\sigma \nabla_z \log p_\sigma(z_\sigma)$$

This formulation allows a deep L -layer network to be partitioned into B independently trainable blocks (e.g., $B = 3$ blocks of 4 layers each), with each block optimized to denoise within a specific noise range.⁴³ Gradients are computed for only one active block per training

step, reducing memory consumption by a factor of B .⁴³

Under highly stochastic reinforcement learning environments (such as navigating a maze with sparse rewards or solving complex combinatorics), combining block-wise independent training with policy gradients introduces unique dynamical challenges:

Gradient Staleness and Representational Coordinate Mismatch

Because only $1/B$ of the network is active during a given step, the parameters of the remaining blocks are frozen.⁴³ The active block updates its parameters to maximize the GDPO objective.³⁹ However, because the upstream blocks are frozen, the distribution of the latent context z arriving at the active block is non-stationary. When the active block finishes its training phase and the optimization focus shifts to a frozen block, the coordinate representation of the latent space may have drifted, resulting in *representational mismatch*.

Preventative Mitigation Strategies

To stabilize the training of DiffusionBlocks in RL loops, three key architectural adaptations must be enforced:

1. **Equi-Probability Partitioning with Shared Objectives:** Rather than partitioning noise levels uniformly, noise-level ranges are assigned to blocks based on *equal cumulative probability mass*.⁴⁴ This ensures that each block confronts an equally challenging learning problem, balancing parameter updates across the network.⁴⁴
2. **EMA Coordinate Anchoring:** The frozen blocks' inputs and outputs are conditioned using an Exponential Moving Average (EMA) of the active block's representation geometry, preventing sudden topological shifts.
3. **Randomized/Stochastic Block Scheduling:** Rather than updating blocks in a strict sequential order (which causes the model to overfit to one block's objective and forget the others), the active block is sampled stochastically at each step from a categorical distribution, preserving global coherence.⁴³

Comparative Architectural Overview

The table below provides a comprehensive comparison of SEF-GRAM with leading model-based reinforcement learning and autoregressive planning architectures.

Feature Dimension	DreamerV3	PlaNet	MuZero	Autoregressive VLM Planners	SEF-GRAM (Ours)
Latent Dynamics Formalism	Recurrent State-Space Model (RSSM) with discrete categorical latents	Continuous-time Recurrent State-Space Model (RSSM)	Deterministic transition model parameterized by a standard feedforward network	Discrete, token-level autoregressive sequencing	Continuous-time exact ODE solution via Error-Free Linear Attention (EFLA) ¹

Observation Reconstruction	High-dimensional pixel reconstruction using a heavy decoder	Full pixel reconstruction via convolutional decoders	Reconstruction-free; reconstructs only the value and reward	Not applicable; operates directly on textual token predictions	Completely reconstruction-free; optimized in latent space via VJEP/BJEPA ⁵
Discretization Errors	High; uses discrete time steps	High; subject to numerical integration drift	Not applicable (discrete step transitions)	High; planning is bounded by token-level sequence discretization	Zero discretization error; infinite-order Runge-Kutta continuous integration ¹
Planning Loop Search Mechanism	Latent-space policy execution via actor-critic rollouts	Latent-space planning using the Cross-Entropy Method (CEM)	Monte Carlo Tree Search (MCTS) guided by a learned value network	Externalized autoregressive rollout (Chain-of-Thought prompting)	Bayesian Product of Experts (PoE) with modular task priors ⁶
Optimization Stack	AdamW; requires backpropagation through the entire network	Standard SGD/AdamW; fully end-to-end backpropagation	Standard end-to-end backpropagation	Standard autoregressive language modeling optimization	Muon Orthogonalizer + DiffusionBlocks block-wise training + GDPO ³⁶
Multi-Objective Alignment	Weighted scalar reward sum; prone	Single scalar objective	Single scalar value function	Direct Preference Optimization (DPO) or	Decoupled group-wise advantage normalization

	to reward imbalance		objective	RLHF/RLAIF	on (GDPO) 39
VRAM Footprint	Large (requires keeping the entire encoder-decoder-world-policy stack in memory)	Large (due to image reconstruction decoders)	Large (due to MCTS search tree expansion and value models)	Massive (typically requires multi-GPU clusters for $>$ parameter models)	Ultra-compact; trained on a single consumer GPU ($<$ VRAM) 44

Potential Failure Modes and Preventative Engineering Solutions

The table below outlines the primary failure modes of the unified SEF-GRAM architecture, complete with detailed preventative engineering solutions:

Component	Failure Mode	Trigger Mechanism	Engineering Solution
EFLA Latent Dynamics	Spectral Radius Divergence ²	Occurs when the update gate β_t scales uncontrollably or when key vectors k_t develop large magnitudes, pushing eigenvalues $e^{-\beta_t \lambda_t} > 2$	1. Implement strict L_2 query-key normalization. ¹³ 2. Clamp step size parameters: $\beta_t \leftarrow \sigma(\beta_{\text{raw}}) \cdot$ ¹¹ 3. Initialize transition weights orthogonally to promote well-conditioned covariance. ¹⁴
EFLA Latent	Floating-Point	Small key norms	1. Utilize a

Dynamics	Underflow ($\alpha_t \rightarrow$ or $0/0$ division) ³	(from padding or dropout) cause $\lambda_t =$, leading to division-by-zero or precision loss in $\alpha_t = \frac{1 - e^{-\lambda_t}}{\lambda_t}$	high-precision Taylor series expansion fallback for $\lambda_t < 1$. ³ 2. Compute exponential decay using the native, stabilized expm1 operator. ⁹
Manifold Alignment	Representation Drift and Exposure Bias ¹⁶	Long-horizon latent planning rollouts drift off the valid encoder manifold due to compounding single-step prediction errors ¹⁶	1. Enforce Theoretical-Latent Hypothesis Encoding (TLHE) with an EMA Target Encoder. ⁵ 2. Regularize the embedding space via Metric Latent Loss to preserve geodesic distances. ²³
Product of Experts	Mean-State Collapse in Non-Convex Landscapes ⁶	Fusing a Gaussian dynamics likelihood with a non-convex, multi-modal obstacle prior shifts the analytical mean μ_{post} directly into the obstacle	1. Detect high-variance prior gradients. 2. Fall back to Energy-Based Langevin Dynamics (SGLD) or Latent Particle Filtering when non-convex boundaries are detected. ⁶
GDPO Optimization	Gradient Underflow in Secondary	Decoupled normalization of a	1. Implement Conditioned

	Objectives ³⁸	highly complex objective (e.g., logical correctness) produces small relative advantages compared to a simple objective (format)	Reward Scheduling (e.g., format reward is strictly zero unless correctness is achieved). ⁴² 2. Apply batch-wise advantage scaling to equalize objective magnitude. ³⁹
DiffusionBlocks	Coordinate Mismatch and Gradient Staleness ⁴³	Updating only $1/B$ of the Transformer blocks per step causes representational drift relative to the frozen, non-stationary upstream layers ⁴³	1. Employ Equi-Probability Partitioning based on cumulative probability mass. ⁴⁴ 2. Anchor latent coordinates using EMA block connections. 3. Implement stochastic, randomized block scheduling. ⁴³

Unexplored Research Avenues and Future Directions

1. Channel-Wise Decoupling via Gated DeltaNet-2 Dynamics

A significant limitation of standard EFLA is its scalar step-size β_t , which forces memory erasure (on the key axis) and memory writing (on the value axis) to scale uniformly across all feature channels.¹⁰ A promising research avenue involves extending the EFLA continuous-time dynamics to incorporate the mathematical formalism of Gated DeltaNet-2.¹⁰

By introducing a channel-wise erase gate $b_t \in d_k$ and a channel-wise write gate $w_t \in d_v$, the model can surgically overwrite specific dimensions of the latent space without affecting neighboring coordinates ¹⁰:

$$S_t = (I - k_t(b_t \odot k_t)^\top) \text{Diag}(\alpha_t) S_{t-1} + k_t(w_t \odot v_t)^\top$$

Deriving the exact continuous-time ODE solution for this channel-wise decoupled formulation would establish a new state-of-the-art in high-fidelity, linear-time latent world models.¹⁰

2. Lie Algebra Homomorphic Latent Transitions

To improve the generalization of the latent planner to complex spatial manifolds (such as higher-dimensional maze routing or 3D robotic arm manipulation), latent transitions can be parameterized using homomorphic updates derived from Lie Algebra.²⁰ Instead of modeling transitions as generic non-linear mappings, forcing the fast-weight transition operators to act as smooth, compositionally consistent transformations in a continuous Lie Group would guarantee that latent planning trajectories satisfy strict geometric invariants (such as translational and rotational equivariance).²⁰ This mathematical constraint would dramatically improve the zero-shot generalization of the planner when transferring from simple 2D mazes to complex, unseen physical topologies.²⁰

3. Second-Order Preconditioning for Latent Regression

Existing delta-rule and fast-weight updates can be interpreted as first-order stochastic gradient steps that optimize a local least-squares regression objective.⁴⁸ However, these updates completely ignore the curvature of the least-squares loss landscape.⁴⁸

A compelling research path is the integration of approximate second-order preconditioning (analogous to Preconditioned DeltaNet) directly into the continuous-time EFLA updates.⁴⁸ By introducing a diagonal approximation of the test-time Hessian matrix, the model can accelerate convergence and resolve information decay, allowing the world model to adapt its transition dynamics to rapid environmental changes in a highly sample-efficient manner.⁴⁸

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